Bayesian Estimation of Co-occurrence *Affinity* With Dyadic Regression

Abstract

Estimating underlying co-occurrence relationships between pairs of species has long been a challenging task in ecology as the extent to which species actually co-occur is partially dependent on their prevalence. While recent work has taken large steps towards solving this problem, the next question is how to assess the factors that influence co-occurrence. Here I show that a recently proposed co-occurrence metric can be improved upon by assigning Bayesian priors to the latent co-occurrence relationships being estimated. In the context of analysing the factors that affect co-occurrence relationships, I demonstrate the need for a generalised linear model (GLM) that takes raw data (co-occurrences and species prevalence) not derived quantities (co-occurrence metrics) as its data. Finally, I show the form that such a GLM should take in order to perform Bayesian inference while accounting for non-independence of dyadic matrix data (e.g. distance and co-occurrence matrices).

Introduction

The analysis of patterns of co-occurrence between taxa is an important and active area of ecological research (Gotelli and McCabe 2002; Barberán et al. 2012; Williams, Howe, and Hofmockel 2014; Kraan, Thrush, and Dormann 2020; Zhu et al. 2023). Mainali et al. (2022) have recently shown that of the numerous ways of measuring co-occurrence relationships between species pairs (reviewed in (Mainali et al. 2022)) the correct and unbiased method is to make use of Fisher's noncentral hypergeometric distribution, as suggested by Veech (2013) and Griffith et al. (2016). Mainali et al. (2022) derived a co-occurrence metric called *affinity* (or $\hat{\alpha}$) based on this distribution. This is a significant step forward in the analysis of co-occurrence relationships. There are still several challenges remaining however. For instance, when species pairs of very low or very high prevalence are analysed with this method they will often be assigned very high or low affinity scores, but with low confidence (high p-value and wide confidence interval). For downstream analysis then the researcher may (i) treat all data points as equal, (ii) remove data above some high p-value threshold or (iii) devise a scheme to weight data appropriately. (i) wastes information and can yield misleading results. (ii) again wastes information and could severely bias results depending on the reasons for the differences in prevalence/p-values. If an appropriate, unbiased scheme for (iii) should be devised, then this would be a welcome development. However, adopting a Bayesian approach that builds upon the work of Mainali et al. (2022) not only yields more accurate estimates of the *affinity* between species, but also naturally propagates uncertainty through the analysis, i.e., it accounts for the differing levels of confidence we have about the co-occurrence relationships between different species pairs. In the present work I illustrate this point and provide the model description and code to use this Bayesian method in practice.

A significant advantage of this framework is that it allows co-occurrence data to be analysed as response data in a Bayesian general linear model (GLM). That is, by supplying species occurrence data as the prevalence of individual species, the number of times a given species pair co-occur and the total number of sites considered, it is now no more complicated to construct a regression model (with a Fisher's noncentral hypergeometric likelihood function) than it would be to perform binomial regression with count data. Similarly, this method makes the best use of all available information and weighs data points appropriately. Thus, just as it is not appropriate to convert count data to proportions and conduct linear regression, it is no longer best practice to summarise co-occurrence data as point estimates for linear regression.

Results

Mainali et al. (2022) show that the log of the odds ratio term in Fisher's noncentral hypergeometric distribution (a quantity they term α) can be used to appropriately describe the extent to which two species will tend to co-occur more or less than would be expected based just on the prevalence of the species. They go on to propose that the maximum likelihood estimate of this parameter $\hat{\alpha}$ or *affinity* should be used as a pairwise co-occurrence metric. However, Maximum likelihood estimates can yield values of positive or negative infinity. This causes difficulties for downstream analyses (one cannot do something as simple as calculating the mean of a set containing infinite values). Furthermore, we know *a priori* that an infinitely large or small affinity is not sensible for most cases of interest to ecologists. Bayesian analysis uses prior knowledge to avoid the estimation of physically or biologically implausible values. Mainali et al. (2022) reassign these infinite estimates an absolute value of $\log(2N^2)$, where N is the total number of sample sites (from an argument made based on the Jeffreys' prior for the beta distribution). While this figure comes from sound argument, there are at least two problems with this approach. Firstly, not all data are treated the same way, i.e., no regularisation is applied to finite affinity estimates, only to these extreme values. Secondly, the value $\log(2N^2)$ is only a function of N, not the species prevalence. Thus, it is not influenced by our actual state of knowledge about the species in question.

To make these ideas more concrete and show the practical implications, I simulated species pairs using the affinity model. For each pair there are $N = 30$ sites they can inhabit, species A has a prevalence mA and species B has prevalence mB. The number of sites at which they co-occur k was drawn from a Fisher's noncentral hypergeometric distribution

$$k \sim \text{fnchypg}(mA, N - mA, mB, e^{\alpha}),$$

with 10 draws per combination of mA and mB for each of 41 different values of α. Then, given the values for N, k, mA and mB I estimated α using two methods. Firstly, I used the original maximum likelihood estimate (MLE) of Mainali et al. (2022). Next I obtained maximum *a posteriori* (MAP) estimates with a Gaussian prior $\text{N}(0, 3)$ for α. Note that these are not strongly regularising priors as when exponentiated in the likelihood function a standard

deviation of $3 \approx 20$ and two standard deviations $6 \approx 403$ which is a very large odds ratio for most applications. Priors used for analysing real data should be chosen after simulation, to demonstrate that they do not bias against feasible parameter values for the specific research. In order to compare the two methods, for each combination of mA and mB I calculated the deviation between the inferred and ground truth α values as the root mean squared error (RMSE).

Figure 1 shows that only when both mA and mB were equal to 15 was the RMSE approximately equivalent between MLE and MAP methods. Whenever one or both species had a high or low prevalence, and particularly as the absolute value of α became larger, the MLE method produced very poor estimates, and the extreme estimates were always the same $\log(2N^2) = 7.496$. By contrast, for the MAP values, the prior provides regularisation which can be overcome by increasing confidence in the data, which is a function of mA and mB. Thus, the models *best guess* when $mA = 15$, $mB = 5$ and $k = 5$ is higher than the equivalent situation when $mB = 1$ and $k = 1$. Neither of these methods is perfect however. When asked for one, a model will give you it's best guess point estimate, but we can make better use of the data we have collected if we can utilise not only the point estimates but also our uncertainty around them.

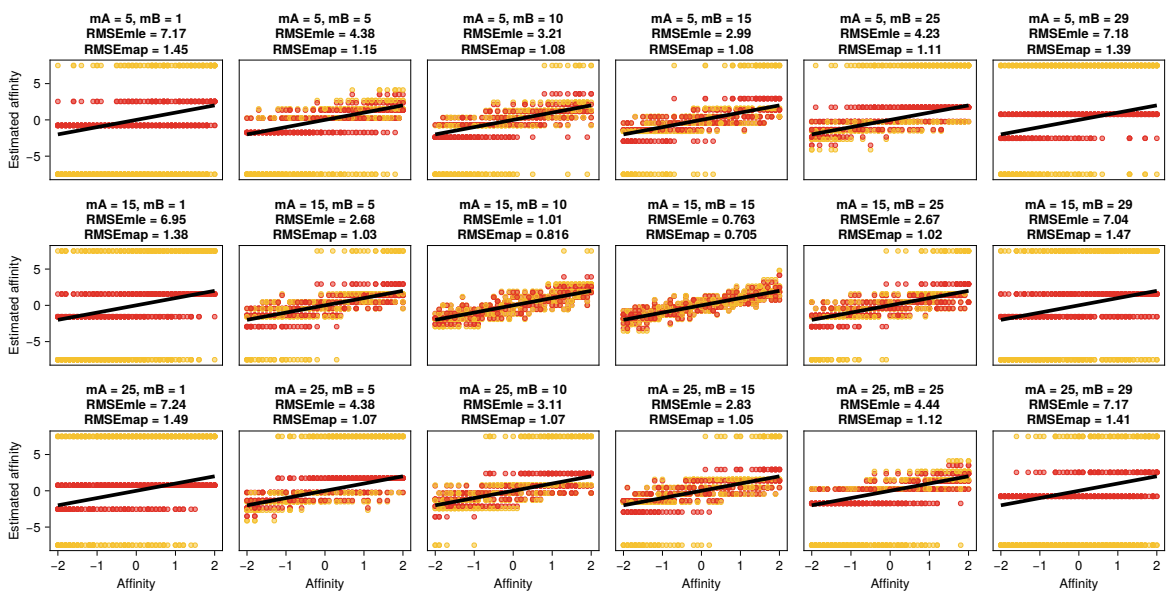

Figure 1: Actual and estimated affinity values for a range of species prevalences. Yellow points are estimates using the original maximum likelihood method and red points are maximum *a posteriori* estimates. Black lines indicate the actual affinity values used to generate the data. In each panel is shown the root mean squared error (RMSE) for both types of estimate.

Often we do not simply wish to report co-occurrence relationships, but to measure how they change with some other variables of interest. For many types of data there are well understood

and regularly used probability distributions which can be used in Bayesian and frequentist GLMs. For co-occurrence data this has not been the case. Given the issues with deriving point estimates highlighted above (Figure 1) it is unlikely that simply fitting a linear model to such point estimates of co-occurrence affinity will yield reliable results. Thus, rather than supplying a second inference model with uninformative point estimates from a previous model, we can provide our regression model with all the data on species prevalence and co-occurrences instead. To demonstrate the impact of this I simulated 41 sets of predictor data $\vec{x}$, each consisting of 30 draws from $\mathrm{N}(0, 1)$. Affinity values were generated by multiplying the predictor data by a regression coefficient β. For each affinity value - species prevalence mA and mB values were chosen randomly between 1 and 29 and a k value was drawn from the Fisher's noncentral hypergeometric distribution as above. For each generated data set pairwise affinity values were estimated by the MAP and MLE methods. Then linear regression analysis was conducted on these point estimates $\alpha \sim \beta\vec{x}$. Additionally I obtained a maximum likelihood estimate of a GLM of the form

$$\begin{aligned} \vec{k}_i &\sim \mathrm{fnchypg}(\vec{mA}_i, N - \vec{mA}_i, \vec{mB}_i, e^{\alpha}) \\ \alpha_i &= \gamma + \beta\vec{x}_i, \end{aligned} \tag{1}$$

where $\vec{k}$, $\vec{mA}$ and $\vec{mB}$ are vectors containing the values of k, mA and mB respectively and γ is the intercept.

The results in Figure 2 show how poorly fitting a linear model to $\hat{\alpha}$ point estimates does, typically overestimating the absolute value of β by a large margin. Using MAP estimates of α does better here, exhibiting the opposite behaviour of slightly underestimating the absolute value of β. However, by cutting out the step of generating point estimates for each pair the GLM retains all pertinent information and accurately recaptures the parameters of the data generating model. It is of course expected that the GLM should be able to discover the correct parameter values, since they were generated by an identical model. What is important is the way the other two models fail by comparison, and of course the fact that we now have the correct likelihood function for such a co-occurrence GLM.

In many if not most cases when working with co-occurrence data it will be in the form of a square co-occurrence matrix similar to the distance and dissimilarity matrices used to record e.g. phylogenetic distances between species or community dissimilarities between sample sites. As with these other types of matrices, if we wish to perform regression analysis treating each entry in the matrix as data point, we must account for non-independence of data coming from the same row or column, e.g. same site, species etc.. To deal with this we include a random effect λ for each species (Clarke, Rothery, and Raybould 2002; Gompert et al. 2014). Thus, whereas the GLM above contains the term

$$\alpha_i = \gamma + \beta\vec{x}_i$$

we would now have

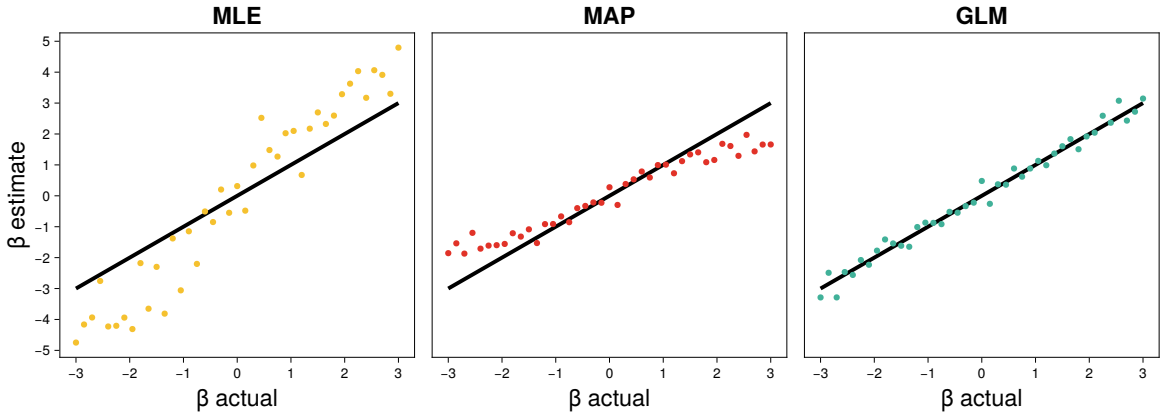

Figure 2: Estimated regression coefficients β according to three different methods: fit linear model to $\hat{\alpha}$ values, fit linear model to MAP estimates of α, fit GLM to raw data. Black lines indicate the true β values.

$$\alpha_i = \lambda_i + \lambda_j + \beta X_{ij}.$$

Where X is a (possibly dissimilarity or distance) matrix in which element X_{ij} is a quantity of interest relating species i to species j. Given a similarly arranged matrix K which holds the k values for all species pairs and a vector $\vec{m}$ containing species prevalence, our dyadic GLM becomes

$$\begin{aligned} K_{ij} &\sim \text{fnchypg}(\vec{m}_i, N - \vec{m}_i, \vec{m}_j, e^{\alpha}) \\ \alpha_{ij} &= \lambda_i + \lambda_j + \beta X_{ij} \end{aligned} \tag{2}$$

for each index ij in either the lower or upper triangle of K. While Maximum likelihood estimates of the λ and β parameters can be obtained, we still need a way to properly account for our uncertainty in our estimates. Bayesian inference provides an intuitive framework for this, grounded in probability theory. Thus, we can construct a Bayesian GLM by assigning priors to the unknown parameters. Assuming Gaussian priors for all parameters we have

$$\begin{aligned} K_{ij} &\sim \text{fnchypg}(\vec{m}_i, N - \vec{m}_i, \vec{m}_j, e^{\alpha}) \\ \alpha_{ij} &= \lambda_i + \lambda_j + \beta X_{ij} \\ \beta &\sim \text{N}(0, \beta_\sigma) \\ \lambda_i &\sim \text{N}(0, \lambda_\sigma) \end{aligned} \tag{3}$$

where β_σ and λ_σ are to be chosen according to the specifics of the system being analysed.

Discussion

Here, I have shown how to construct a Bayesian dyadic GLM for the analysis of co-occurrence data. This builds on the work of Mainali et al. (2022) as well as Veech (2013) and Griffith et al. (2016). The identification of Fisher's noncentral hypergeometric distribution (or mathematically equivalent formulations) as the correct distribution for modelling co-occurrence led first to null model approaches to co-occurrence analysis (Griffith, Veech, and Marsh 2016; Veech 2013), then to a useful co-occurrence metric (Mainali et al. 2022) and now to a general model capable of analysing raw co-occurrence data as a response variable even when data points are not independent (which will generally be the case). It should be noted that the co-occurrence relationships discussed here and in the works cited above are probabilistic in nature, i.e., I do not assume that either a high or low affinity between a pair of species implies significant ecological interaction.

Part of the motivation for this work was the failure to recapture known regression coefficients when fitting linear models to pairwise affinity estimates (Figure 2). There may be other ways of combating this failure. For example, the removal of data points for which we have low confidence may be an option. For instance if $N = 30$, $mA = 29$ and $mB = 1$, then a k of 1 tells us very little, since our null expectation is that k will very likely $= 1$. This will lead to a high affinity estimate but with a wide confidence interval and a high p-value. Using a cut-off threshold e.g. only using data for which $p < 0.05$ may lead to better results. However, these data points are now *missing* from further analyses. The potential pitfalls involved in dealing with missing data are numerous, nuanced and beyond the scope of the present work (Kang 2013). It is unnecessary to risk the possible bias associated with systematically removing data points when the Bayesian analysis framework naturally accounts for differing levels of confidence between data (McElreath 2018).

The method proposed here provides ecologist with an important new tool for the analysis of co-occurrence, and in particular discovering the relationships between co-occurrence and other variables e.g. phylogenetic distance, which is an active area of research (Goberna et al. 2019) and has been the subject of much research effort over the past few decades (Webb et al. 2002) but only now has an analysis framework based on the simple application of probability theory (McElreath 2018) with correct modelling of co-occurrence probabilities (Griffith, Veech, and Marsh 2016; Veech 2013; Mainali et al. 2022) while accounting for non-independence of data in matrices of pairwise species measurements (Clarke, Rothery, and Raybould 2002; Gompert et al. 2014).

Methods

Maximum likelihood estimates of co-occurrence affinity were obtained using the R (R Core Team 2014) package `co-occurrenceAffinity` (Mainali et al. 2022). All other analyses and visualisations were carried out in Julia (Bezanson et al. 2017). All models were constructed in the probabilistic programming language `Turing` (Ge, Xu, and Ghahramani 2018) with MLE

estimates of regression coefficients fit to data using the Nelder-Mead method (Nelder and Mead 1965) and MAP estimates of α fit using L-BFGS (Liu and Nocedal 1989) implemented in `Optim` (Mogensen and Riseth 2018) and results visualised in `Makie` (Danisch and Krumbiegel 2021).

Code availability

All code to produce the figures and the manuscript can be found at

https://github.com/EvoArt/bayesian-affinity.

Model implementations in Turing

Bayesian GLM

$$
\begin{aligned}
\vec{k}_i &\sim \text{fnchypg}(\vec{mA}_i, N - \vec{mA}_i, \vec{mB}_i, e^{\alpha}) \\
\alpha_i &= \gamma + \beta \vec{x}_i \\
\beta &\sim \text{N}(0, \beta_{\sigma}) \\
\gamma &\sim \text{N}(0, \beta_{\gamma})
\end{aligned}
$$

```
@model function reg(x, N, mA, mB, k, priors)   ~ Normal(0,prior[1])   ~ Normal(0,prior[2])
  for i in eachindex(x)
        =   +  *x[i]
      k[i] ~ FisherNoncentralHypergeometric(mA[i],N-mA[i],mB[i],exp( ))
  end
end
```

Dyadic Bayesian GLM

$$
\begin{aligned}
K_{ij} &\sim \text{fnchypg}(\vec{m}_i, N - \vec{m}_i, mj, e^{\alpha}) \\
\alpha_{ij} &= \lambda_i + \lambda_j + \beta X_{ij} \\
\beta &\sim \text{N}(0, \beta_{\sigma}) \\
\lambda_i &\sim \text{N}(0, \lambda_{\sigma})
\end{aligned}
$$

```
@model function dyadic_glm(X, N, m, K, priors)
    n = size(k,1)
      ~ Normal(0,prior[1])
      ~ filldist(Normal(0,prior[2]),n)

    for j in 1:n-1
```

```
        mB = m[j]
        for i in j+1:n
            mA = m[i]
              =   *X[i,j] + [i] + [j]
            K[i,j] ~ FisherNoncentralHypergeometric(mA,N-mA,mB,exp( ))
        end
    end
end
```


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